

Ranadeep Daw

Curriculum Vitae

Department of Mathematics & Statistics
University of West Florida
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Current Position

Aug 2024– Assistant Professor of Statistics, University of West Florida

Research Interests

Spatiotemporal modeling, low-rank methodologies, data reduction, environmental risk, random neural networks, computational tools, Bayesian analysis

Education

2018–2023 **Ph.D. in Statistics**, *University of Missouri - Columbia*
GPA: 3.878

Advisor: Dr. Christopher K. Wikle

Thesis topic: Methodologies for low-rank analysis and regionalization for multi-scale spatial datasets

2014–2016 **Master of Statistics (M. Stat.)**, *Indian Statistical Institute*, Kolkata

2011–2014 **Bachelor of Statistics (B. Stat.)**, *Indian Statistical Institute*, Kolkata

Work Experience

Jul 2023– Jul 2024 **Postdoctoral fellow**, *National Institute of Environmental Health Sciences*, Durham, NC
Working on spatiotemporal exposure modeling and supervised environmental risk modeling.

Aug 2018– Jul 2019 **Intern**, *Missouri Department of Conservation*, Columbia, MO
Served as part of my research assistantship. Worked on species distribution models, small area estimation, and bootstrapping.

Jun 2016– **Data Scientist**, *Deloitte US India*, Hyderabad, India

Dec 2017 Addressed statistical queries in various business-related sectors and implement them in user-friendly tools for third-party use. Worked on real-world data from the insurance, banking, medical, and retail sectors.

May 2015– Jul 2015 **Intern**, *General Electric*, Bangalore, India
Implemented fast heuristic time series algorithms.

May 2014– Jul 2014 **Intern**, *FinIQ*, Pune, India
Visual Basic-based automation of financial algorithms in Excel.

Research Projects

Accepted Papers

- 1 Yeasmin, F., **Daw, R.**, Chakraborty, B., Gupta, A., Bhattacharya, S., Chakraborty, B. (2021). A new growth rate measure in identifying extended Gompertz growth curve and development of goodness-of-fit test. *Calcutta Statistical Association Bulletin*, Volume 73, 127–145.
- 2 Chakraborty, S., Menifield, C. E., **Daw, R.** (2022). Impact of Stand Your Ground, Background Checks, and Conceal and Carry Laws on homicide rates in the US. *Journal of Public Management and Social Policy*.
- 3 **Daw, R.**, Simpson, M., Wikle, C. K., Holan, S. H., Bradley, J. R. (2022). An overview of univariate and multivariate Karhunen Loève expansions in Statistics. *Journal of the Indian Society for Probability and Statistics*, 1–42.
- 4 **Daw, R.**, Wikle, C. K (2022). REDS: Random Ensemble Deep Spatial prediction. *Environmetrics*, e2780, 1180–4009, DOI 10.1002/env.2780.

- 5 **Daw, R.**, Wikle, C. K. (2022). Supervised spatial regionalization using the Karhunen-Loève Expansion and minimum spanning tree. *Journal of Data Science* 20 no. 4. 566–584, DOI: 10.6339/22-JDS1077.
- 6 Schliep, E. M., Wikle, C. K., **Daw, R.** (2023). Correcting for informative sampling in spatial covariance estimation and kriging predictions. *Journal of Geographical Systems*, 1–27.
- 7 **Daw, R.**, Wikle, C. K. (2025+). Efficient computation of orthogonal basis functions for spatial change-of-support problems. Accepted at *Gujarat Journal of Statistics and Data Science*.
- 8 **Daw, R.**, Bagui, S. C. (2026). Teaching Bayesian Modeling to Beginners: A Case Study Using Educational Data. *International Journal of Science and Research*, Volume 15 Issue 5, DOI: <https://dx.doi.org/10.21275/SR26502014614>.

Under Revision

- 9 **Daw, R.**, Bhattacharya, I., Chakraborty, S. (2026+). Causal Effects of Urban Amenities on airbnb Prices: A Hybrid Spatial Filtering Approach. Under revision at *Statistical Analysis and Data Mining*.
- 10 **Daw, R.**, Evans, H., Bhattacharya, I. (2026+). Socioeconomic Drivers of Physical Morbidity Across U.S. Counties: A Spatial Causal Inference Approach. Under revision at *Journal of Geographical Systems*.

Submitted Papers

- 11 **Daw, R.**, Bradley, J. R., Wikle, C.K., Holan S. H. (2025+). A Criterion for Aggregation Error for Multivariate Spatial Data. Submitted to *ASA Discoveries*.
- 12 **Daw, R.** (2026+). Spatial Coreset Construction for County-Health Outcomes via Spectral Basis Regression. Submitted to *Journal of Biomedical and Health Informatics*.

Ongoing Papers

- 13 **Daw, R.** (2026+). Analysis of McKinney-Vento Requirements in Florida: a machine learning approach

Conferences

Invited Talks

- 2025 **FL-ASA**, (*Florida ASA Chapter*), Causal Effects of Urban Amenities on airbnb Prices: A Hybrid Spatial Filtering Approach.
- 2025 **JMM**, (*Joint Mathematics Meetings*), Efficient computation of orthogonal basis functions for spatial change-of-support problems.
- 2021 **JSM**, (*Joint Statistical Meetings*), Application of coresets in Spatial Modeling.

Contributed Talks

- 2026 **MAA-FL**, (*Mathematical Association of America - Florida Section*), Application of coresets in Spatial Modeling.
- 2021 **IISA**, (*International Indian Statistical Association*), Application of coresets in Spatial Modeling.
- 2020 **JSM**, (*Joint Statistical Meetings*), Uncertainty Quantification and Inference for Spatiotemporal Forecasting via Echo State Mixture Density Networks with Relevance Propagation.

Poster Presentations

- 2024 **Theory and Foundations**, *Florida State University*, SBoost: A spatially-aware Boosting Model for Large Spatial Prediction.
- 2024 **Science Day**, *NIEHS (National Institute of Environmental Health Sciences)*, SBoost: A spatially-aware Boosting Model for Large Spatial Prediction.
- 2022 **Applied Statistics Symposium**, *ICSA (International Chinese Statistical Association)*, REDS: Random Ensemble Deep Spatial prediction.
- 2019 **Deep Learning Program Opening Workshop**, *SAMSI (Statistical and Applied Mathematical Sciences Institute)*, Deep Neural Network in Cusp Catastrophe Model

2018 **Innovations in Design, Analysis, and Dissemination**, *Frontiers in Biostatistical Methods*, Does Changing Handgun Laws Affect Crime Rates?

Teaching Experience

- Fall 2024 – **Assistant Professor, Statistics, UWF**
current
 - STA4173 Biostatistics
 - STA5176 Statistical modeling
 - STA6235 Regression modeling
- Fall 2019 – **Graduate Instructor, Mizzou**
Spring 2020
 - Stat 3500: Probability and Statistics - II
- Spring 2018 **Teaching Assistant**, Mizzou

Mentoring Experience

- 1 Supervised six undergraduate mathematics proseminar projects, including three successful completions and three currently in progress.
- 2 Worked as consultant in the Data Fest, a data analysis workshop organized by the Department of Statistics, University of Missouri.
- 3 Responsible for mentoring two newly-hired employees at Deloitte. Worked on a coding project with one mentee.

Academic Awards

- 2011–2016 Received the **INSPIRE** scholarship administered by the Department of Science and Technology of the Government of India.
- 2010 Cleared the **Regional Mathematical Olympiad** and qualified for the national level examination.

Service & Membership

- University Part of Academic Standards Committee and Proseminar Committee at University of West Florida.
Service Proposal reviewer for OUR Project Award Program.
- Peer Review Reviewer for *Spatial Statistics*, *Signal Processing*, *Environmetrics*, *Data Science in Science*, *Biometrics*, *American Journal of Mathematical and Management Sciences*.
- Professional Member of *International Indian Statistical Association*.
Membership

Technical Expertise

- Computing MATLAB, Python, SAS, R, Julia.
Programming C, C++, L^AT_EX.
Visualization Tableau, HTML, JavaScript, JavaScript-D3.

I hereby do certify that all the above information is true to the best of my knowledge.

Ranadeep Daw
May 7, 2026